



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **RENEWCOILS ENGINEERING AND SUPPLY CO. LTD.**

WPS No. **RNC-WPS-023**

Date **10-Jun-13**

Revision No. **0**

Supporting PQR No.(s) **RNC-PQR-023**

Date **14-May-13**

Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding + Shield Metal Arc welding**

(GTAW + SMAW)

JOINT (QW-402)

Joint Design **Groove/Fillet**

Backing (Yes) **SMAW** (No) **X**

Backing Material (Type) **Weld Metal**

☒ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **8** Group No.

1 to P-No. **8**

Group No. **1**

Material Specification Type and Grade

SA 312 Gr.TP 321/TP347/SA358TP321

SA 312 Gr.TP 321/TP347/SA358TP321

Chem. Analysis and Mech. Prop.

SA403 Gr.WP321/WP347 (as per P-no.8)

TO SA403 Gr.WP321/WP347 (as per P-no.8)

Thickness Range :

Base Metal : Groove **5 - 64 mm.** Fillet **ALL**
Pipe Dia. Range : Groove **ALL** Fillet **ALL**
Other

FILLER METALS (QW-404)

Spec. No. (SFA)

GTAW

SMAW

5.9

5.4

AWS No. (Class)

ER 347 (Solid)

E347-17

F-No.

6

5

A-No.

8

8

Size of Filler Metals

Ø 2.0 & 2.4 mm

Ø 2.6, 3.2, 4.0 mm.

Deposited Weld Metals

5 mm.

27 mm.

Thickness Range :

Groove

10 mm.

54 mm.

Fillet

All

All

Electrode-Flux (Class)

-

Flux Trade Name

-

Consumable Insert

-

Other

Bohler SAS 2-IG or Equivalent

Bohler FOX SA 2-A or Equivalent

POSITION (QW-405)

Position(s) of Groove

ALL

Welding Progression

Up **X** Down

Position(s) of Fillet

ALL

Temperature Range **1080°C ±10°C**

Time Range **Holding Time 1 Hour**

Notes **Heating Rate = 180-200 °C & Air Cooling**

PREHEAT (QW-406)

Preheat Temp.

Min.

30°C (Ambient)

Interpass Temp.

Max.

170°C

Preheat Maintenance

N/A

Notes

N/A

GAS (QW-408)

Percent Composition

	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	10-20 L/Min
Trailing	-	-	-
Backing	Argon	99.99%	10-20 L/Min



WELDING PROCEDURE SPECIFICATION (WPS)

WPS No.

WPS-023

Rev.

0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☒ AC ☒ DC Polarity See Below
Amps (Range) See Below Volts (Range) See Below
Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
(Pure Tungsten, 2% Thoriated, etc.)
Mode of Metal Transfer for GMAW N/A
(Spray arc, short circuiting arc, etc.)
Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS
Orifice or Gas Cup Size 8.0 mm - 19.0 mm
Initial and Interpass Cleaning (Brushing, Grinding, etc.) BOTH APPLICATIONS
Method of Back Gouging N/A
Oscillation N/A
Contact Tube to Work Distance N/A
Multiple or Single Pass (per side) MULTIPASS
Multiple or Single Electrodes SINGLE ELECTRODE
Travel Speed (Range) See Below
Peening N/A
Other

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER347	2 & 2.4 mm.	DCEN	80-120	10-15	3-7	
2nd	GTAW	ER347	2 & 2.4 mm.	DCEN	80-120	10-15	3-7	
3 & Other	SMAW	E347-17	2.6, 3.2, 4.0 mm.	DCEP	80-150	20-30	4-10	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of ASME IX, 2010 latest Revision.

Prepared by:

Mr. Kittipong Meijani
Name

Signed

10 Jun 13

Date

Reviewed by:

ภูชิน สีหอมไชย
Phuchin Seehomchai
Name

Signed

12/06/13

Date

Approved by:

ภูชิน สีหอมไชย
Phuchin Seehomchai
Name

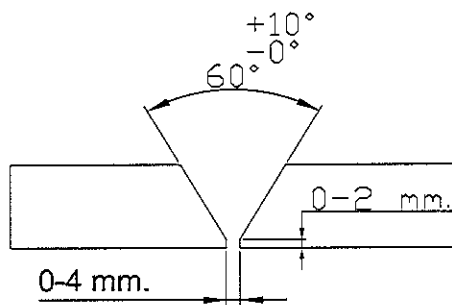
Signed

12/06/13

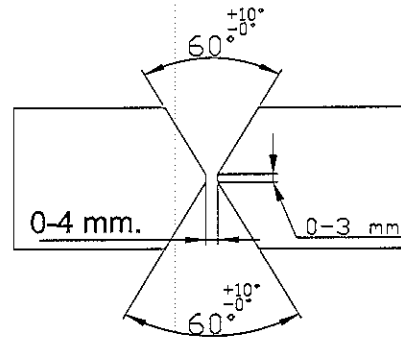
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JOINT PREPARATIONS

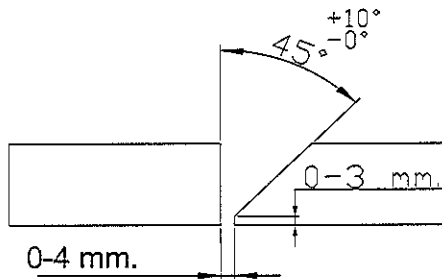
1. Single Vee Groove



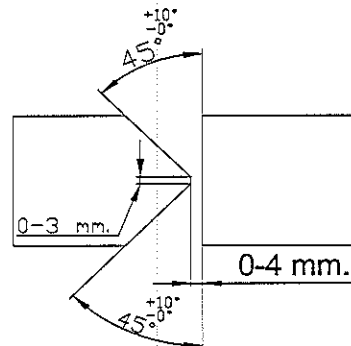
2. Double Vee Groove



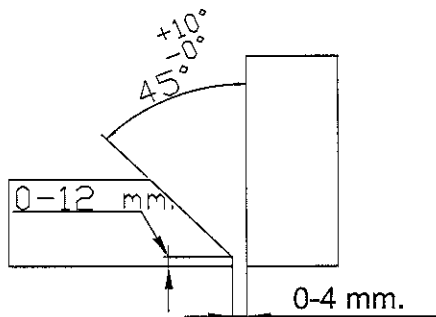
3. Single Bevel Groove



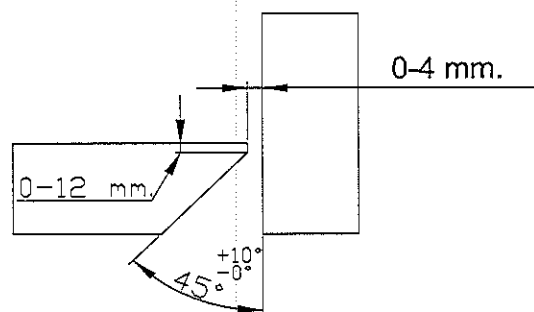
4. Double Bevel Groove



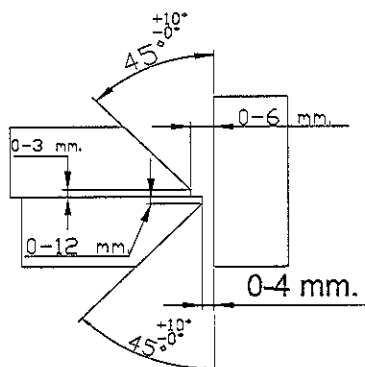
5. Bevel Groove



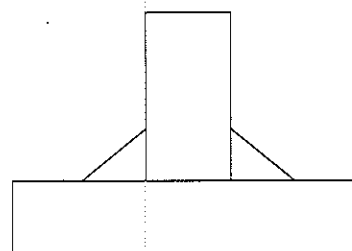
6. Bevel Groove



7. Bevel Groove



8. Fillet





Procedure Qualification Record (PQR)

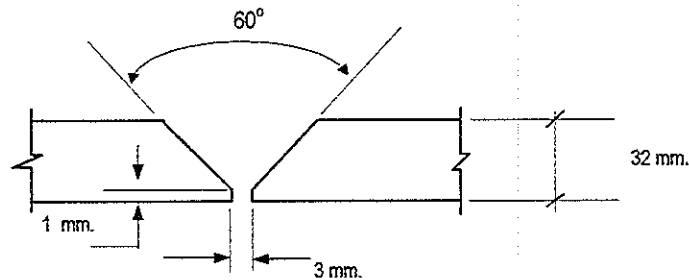
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-023

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-023	Date :	14 May 2013
WPS Number :	RNC-WPS-023		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)			
Material Spec.	SA 312	To	SA 312	Temperature	1080° C		
Type or Grade	TP321	To	TP321	Hold Time	1 HR.		
P. No.	8	To	8	Others	Heating Rate 180° C/Hr Air Cooling		
Thickness of Test Coupon		32 mm.		GAS (QW-408)			
Diameter of Test Coupon		Ø 10"					
Others				Percent Composition			
FILLER METALS (QW-404)					Gas (es)	Mixture	Flow Rate
				Shielding	Argon	99.95%	15 L/Min.
				Trailing	-	-	-
				Backing	Argon	99.95%	17 L/Min.
SFA Specification		GTAW	A 5.9	ELECTRICAL CHARACTERISTICS (QW-409)			
AWS Classification			E 347	Current	DC (GTAW)	DC (SMAW)	
Filler Metal F-No.		6	5	Polarity	EN	EP	
Weld Metal Analysis A-No.		8	8	Amperage	102 - 119 (GTAW)	Voltage	11 - 12 (GTAW)
Size of Filler Metal		Ø 2.4 mm.	Ø 2.6, 4 mm.	119 - 137 (SMAW) 24 (SMAW)			
Others		TGS-347	Bohler FOX SA 2-A	Tungsten Electrode Size	Ø 2.4 mm. 2% thoriated.EWTh-2		
Weld Metal Thickness		-		Others	N/A		
POSITION (QW-405)				TECHNIQUE (QW-410)			
Position of Groove		6G		Travel Speed	5.0 - 6.5 cm/min.		
Progression of Welding (Uphill or Down)			Up Hill	String or Weave Bead	Both		
Others		N/A		Oscillation	N/A		
PREHEAT (QW-406)				Multiple or single Pass		Multiple Pass	
Preheat Temperature		36° C	Postheat Temp.	Multiple or Single Electrodes		Multiple Electrodes	
Interpass Temperature (max.)		161° C		Others		N/A	
Postheat Time		1 Hr.					

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- RNC-PQR-023

Tensile Test Report No.TS-13-05-090

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
1474/13 TR1	19.35	29.49	570.631	346.370	606.995	OUT OF WELD
1474/13 TR2	19.09	29.85	569.837	295.180	518.008	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-05-009

Type and Figure Number	Type of Bend	Indication	Results
1474/13 BS1	Transverse face bend	No Open Discontinuity	Accepted
1474/13 BS2	Transverse face bend	No Open Discontinuity	Accepted
1474/13 BS3	Transverse root bend	No Open Discontinuity	Accepted
1474/13 BS4	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170) Report No.IM-13-05-061

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Average (Joules)	Lateral Expansion Mils
1474/13-1	Weld Metal	V-Notch	- 20 °C	117.9	117.15	
1474/13-2	Weld Metal	V-Notch	- 20 °C	121.49		
1474/13-3	Weld Metal	V-Notch	- 20 °C	112.06		
1474/13-1	HAZ	V-Notch	- 20 °C	103.91	125.85	
1474/13-2	HAZ	V-Notch	- 20 °C	127.39		
1474/13-3	HAZ	V-Notch	- 20 °C	146.26		
1474/13-1	Base Metal	V-Notch	- 20 °C	272.92	260.91	
1474/13-2	Base Metal	V-Notch	- 20 °C	254.66		
1474/13-3	Base Metal	V-Notch	- 20 °C	255.16		

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR-014
Hardness Testing Report No. HV-13-05-042
Macrostructure Report No. MA-13-05-067

Welders Name Mr.Somjai S. / Mr.Kaen H. Welder Number W-081 / W-079
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

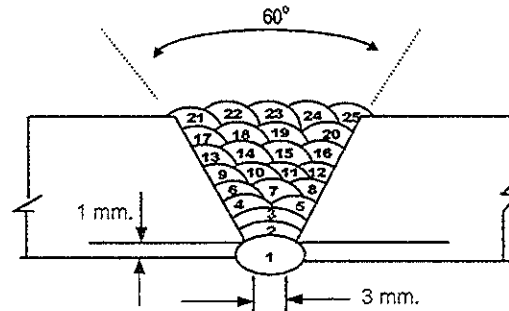
Manufacture or Contractor : PAE Technical Service Company Limited Inspected by : Mr.Sayam S.
Date : 08/06/13 Certified by : Mr.Chaloemkiet R.
Date : 7 June 2013

PAE Technical Service Company Limited

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (662) 721-2577

Public Registration No. (5) 1571/2542

Welding Sequences for RNC-PQR-023 (Butt Joint)



Ø10" T.32 mm

Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER347	2.4	102	11	DCEN	-	-	5.50	12.20	15	17
2	GTAW	ER347	2.4	119	12	DCEN	-	-	5.00	17.10	15	17
3	SMAW	E347-16	2.6	120	24	DCEP	-	-	5.50	31.40	-	-
4	SMAW	E347-16	2.6	122	24	DCEP	-	-	6.00	29.30	-	-
5	SMAW	E347-16	2.6	120	24	DCEP	-	-	6.00	28.80	-	-
6	SMAW	E347-16	2.6	121	24	DCEP	-	-	6.50	26.80	-	-
7	SMAW	E347-16	2.6	119	24	DCEP	-	-	5.50	31.20	-	-
8	SMAW	E347-16	2.6	121	24	DCEP	-	-	5.50	31.70	-	-
9	SMAW	E347-16	4	134	24	DCEP	-	-	6.00	32.20	-	-
10	SMAW	E347-16	4	134	24	DCEP	-	-	6.00	32.20	-	-
11	SMAW	E347-16	4	135	24	DCEP	-	-	6.00	32.40	-	-
12	SMAW	E347-16	4	136	24	DCEP	-	-	6.50	30.10	-	-
13	SMAW	E347-16	4	135	24	DCEP	-	-	6.00	32.40	-	-
14	SMAW	E347-16	4	136	24	DCEP	-	-	6.50	30.10	-	-
15	SMAW	E347-16	4	136	24	DCEP	-	-	6.50	30.10	-	-
16	SMAW	E347-16	4	136	24	DCEP	-	-	6.00	32.60	-	-
17	SMAW	E347-16	4	137	24	DCEP	-	-	5.50	35.90	-	-
18	SMAW	E347-16	4	136	24	DCEP	-	-	6.00	32.60	-	-
19	SMAW	E347-16	4	136	24	DCEP	-	-	5.50	35.60	-	-
20	SMAW	E347-16	4	135	24	DCEP	-	-	6.00	32.40	-	-
21	SMAW	E347-16	2.6	121	24	DCEP	-	-	5.00	34.80	-	-
22	SMAW	E347-16	2.6	122	24	DCEP	-	-	5.50	31.90	-	-



PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT
 69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG, BANGKOK 10250
 Tel : 02) 322-0222, Fax : 02) 721-2577, RAYONG (038) 682208-9

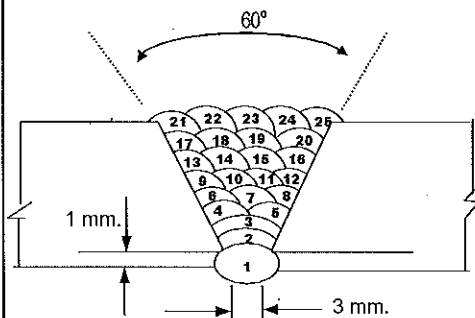
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 3

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : MINI SHUTDOWN PTTGC		Location : RNC Work Shop		Report No. PQR-011	
		Customer Order No. : N/A		WPS No. : RNC-WPS-023		Date : 14-May-13	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
W-081	MR.SOMJAI S.	GTAW	6G	ASME IX	SA 312 Gr. TP321	ø 10" Thk. 32 mm.	A5.9	ACCEPTED	ACCEPTED
W-079	MR.KAEN H.	SMAW					A5.4		

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER 347	2.4	DCEN	102	111	5.5	36	12.2	Argon	15	Argon	17
2	GTAW	ER 347	2.4	DCEN	119	12	5.0	124	17.1	Argon	15	Argon	17
3	SMAW	E347-16	2.6	DCEP	120	24	5.5	161	31.4	-	-	-	-
4	SMAW	E347-16	2.6	DCEP	122	24	6.0	152	29.3	-	-	-	-
5	SMAW	E347-16	2.6	DCEP	120	24	6.0	154	28.8	-	-	-	-
6	SMAW	E347-16	2.6	DCEP	121	24	6.5	150	26.8	-	-	-	-
7	SMAW	E347-16	2.6	DCEP	119	24	5.5	134	31.2	-	-	-	-
8	SMAW	E347-16	2.6	DCEP	121	24	5.5	135	31.7	-	-	-	-
9	SMAW	E347-16	4	DCEP	134	24	6.0	140	32.2	-	-	-	-
10	SMAW	E347-16	4	DCEP	134	24	6.0	154	32.2	-	-	-	-

PAE Inspection By

SIGNATURE :

NAME :

DATE :

Mr. Sayam S.

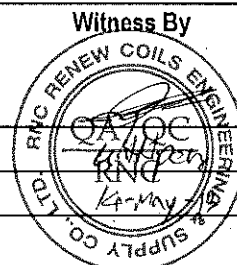
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Witness By

SIGNATURE :

NAME :

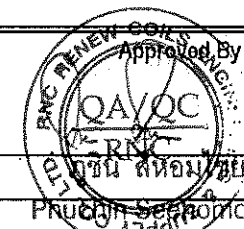
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SIGNATURE :

NAME :

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12/06/13



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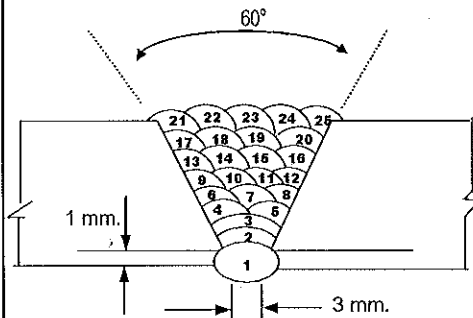
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 2 of 3

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : MINI SHUTDOWN PTTGC		Location : RNC Work Shop		Report No. PQR-011	
		Customer Order No. : N/A		WPS No. : RNC-WPS-023		Date : 14-May-13	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
W-081	MR.SOMJAI S.	GTAW	6G	ASME IX	SA 312 Gr. TP321	ø 10" Thk. 32 mm.	A5.9	ACCEPTED	ACCEPTED
W-079	MR.KAEN H.	SMAW					A5.4		

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
11	SMAW	E347-16	4	DCEP	135	24	6.0	154	32.4	-	-	-	-
12	SMAW	E347-16	4	DCEP	136	24	6.5	150	30.1	-	-	-	-
13	SMAW	E347-16	4	DCEP	135	24	6.0	150	32.4	-	-	-	-
14	SMAW	E347-16	4	DCEP	136	24	6.5	154	30.1	-	-	-	-
15	SMAW	E347-16	4	DCEP	136	24	6.5	150	30.1	-	-	-	-
16	SMAW	E347-16	4	DCEP	136	24	6.0	150	32.6	-	-	-	-
17	SMAW	E347-16	4	DCEP	137	24	5.5	151	35.9	-	-	-	-
18	SMAW	E347-16	4	DCEP	136	24	6.0	152	32.6	-	-	-	-
19	SMAW	E347-16	4	DCEP	136	24	5.5	151	35.6	-	-	-	-
20	SMAW	E347-16	4	DCEP	135	24	6.0	151	32.4	-	-	-	-

PAE Inspection By

SIGNATURE :

NAME :

DATE :

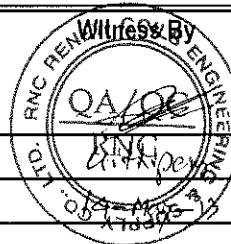
Mr.Sayam S.

14-May-2013

SIGNATURE :

NAME :

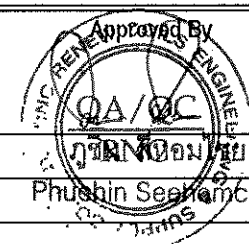
DATE :



SIGNATURE :

NAME :

DATE :



Phuchin Seemachai



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail: qa@lpnplm.co.th



584

TESTING
No. 0154

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-05-090

Revision No. 0

Test Product	PQR No. RNC-PQR-023	Material Characterization	Pipe
Project Name	MINI SHUTDOWN PTGCG	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	SA312 TP321	Type of Test	Reduced Section Tension
Dimension	Ø 10" (32 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	23 May 2013	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	28 May 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	1474/13 TR1	N/A	29.49	19.35	570.631	N/A	N/A	N/A	346.370	606.995	N/A	OUT OF WELD
2	1474/13 TR2	N/A	29.85	19.09	569.837	N/A	N/A	N/A	295.180	518.008	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-023

Welder Name : MR.SOMJAI S./MR.KAEN H.

Welding Process / Position : GTAW+SMAW / 6G

Welder No. : W-081 / W-079

Filler/Electrode Spec't : ER 347+E347-16

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 07 / 06 / 13

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 07 / 06 / 13

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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4. This report is tested for LPN Metallurgical Research Center (Thailand) Co.,Ltd.



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th



Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-05-090

Revision No. 0

Test Product	PQR No. RNC-PQR-023	Material Characterization	Pipe
Project Name	MINI SHUTDOWN PTTC	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	SA312 TP321	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 10" (32 mm Thk.)	Diameter of Meudrel	40 mm
Date Received	23 May 2013	Bend Angle	180 Degree.
Date Tested	28 May 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	1474/13 BS1	Transverse side bend	10.04x28.19x161.79 mm	No Open Discontinuity	N/A	-
2	1474/13 BS2	Transverse side bend	10.00x26.60x161.61 mm	No Open Discontinuity	N/A	-
3	1474/13 BS3	Transverse side bend	9.92x28.13x163.23 mm	No Open Discontinuity	N/A	-
4	1474/13 BS4	Transverse side bend	9.90x28.15x161.77 mm	No Open Discontinuity	N/A	-

Additional details : WPS No. RNC-WPS-023

Welder Name : MR.SOMJAI S./MR.KAEN H.

Welding Process / Position : GTAW+SMAW / 6G

Welder No. : W-081 / W-079

Filler/Electrode Spec't : ER 347+E347-16

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 07 / 06 / 13

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 07 / 06 / 13

LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th



582

TESTING
No. 0130

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-05-061

Revision No. 0

Test Product	PQR No. RNC-PQR-023	Material Characterization	Pipe
Project Name	MINI SHUTDOWN PITGC	Test Method	ASTM A370-10 and ASTM E23M-07a ^{cl}
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	SA312 TP321	Type of Test	Charpy V-Notch Impact
Dimension	Ø 10" (32 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	23 May 2013	Equipment 's Capacity	542 Joules
Date Tested	30 May 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	1474/13-1	-20°C	10.002	10.013	54.94	117.90	117.15	Weld Metal	-
2	1474/13-2	-20°C	10.009	10.015	54.97	121.49			
3	1474/13-3	-20°C	10.011	10.021	54.96	112.06			
4	1474/13-1	-20°C	10.011	10.015	54.92	103.91	125.85	HAZ	-
5	1474/13-2	-20°C	10.009	10.016	54.92	127.39			
6	1474/13-3	-20°C	10.014	10.019	54.93	146.26			
7	1474/13-1	-20°C	10.017	10.013	54.89	272.92	260.91	Base Metal	-
8	1474/13-2	-20°C	10.014	10.017	54.85	254.66			
9	1474/13-3	-20°C	10.007	10.012	54.86	255.16			

Additional details : WPS No. RNC-WPS-023

Welder Name : MR.SOMJAI S./MR.KAEN H.

Welding Process / Position : GTAW+SMAW / 6G

Welder No. : W-081 / W-079

Filler/Electrode Spec't : ER 347+E347-16

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 27 / 06 / 13

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 27 / 06 / 13

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LPN PLATE MILL



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Macrostructure Analysis Report

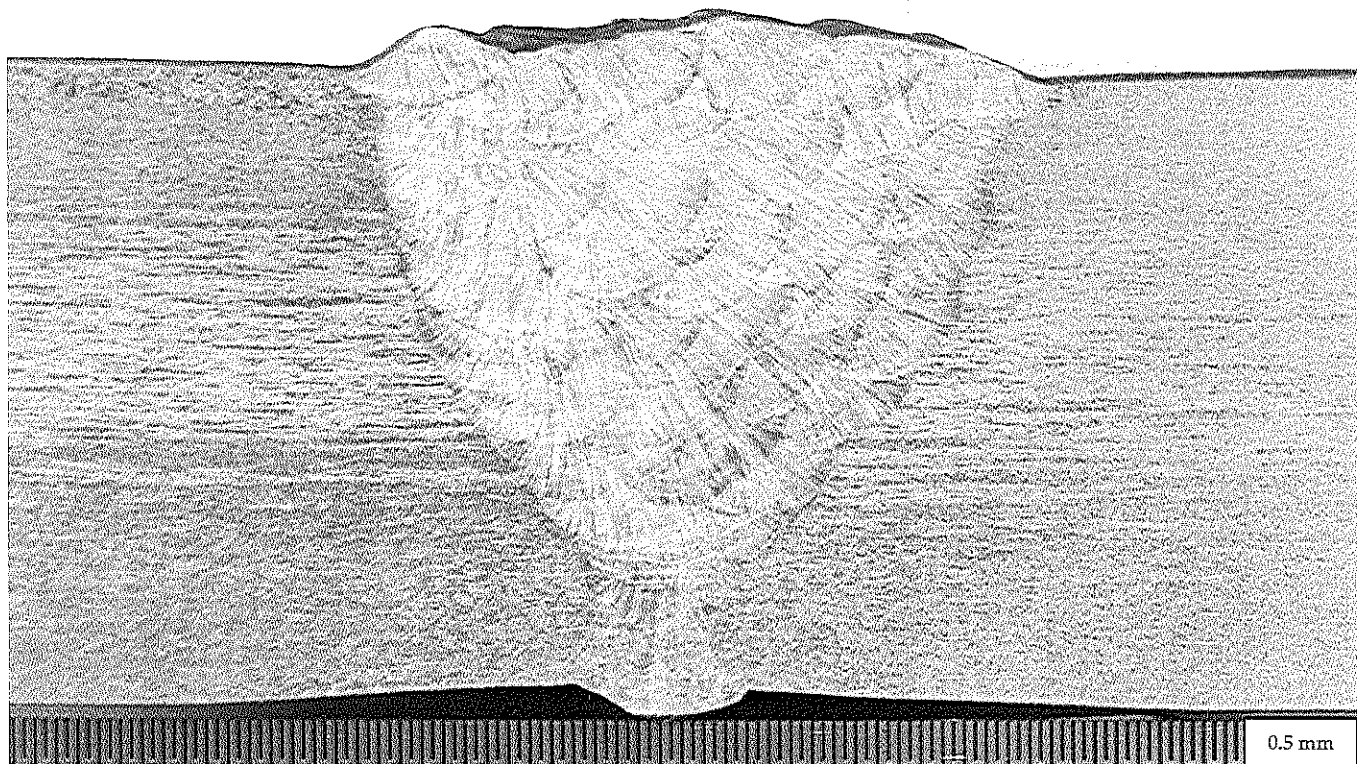
Page 1 of 1

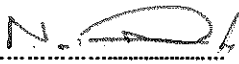
Report No. MA-13-05-067

Revision No.0

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.**Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.**

Test Product	PQR No. RNC-PQR-023	Material Characterization	Pipe
Test No.	1474/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	SA312 TP321	Method of Preparation	ASME Secction IX : 2010 edition
Dimension	Ø10" (32 mm Thk.)	Test Method	ASME Secction IX : 2010 edition
Welding Process/Position	GTAW+SMAW/6G	Standard of Test	ASME Secction IX : 2010 edition
Type of Joint	Butt Joint	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name / No.	MR. SOMJAI S.,MR. KAEN H./W-081,W-079	Time of Etching	5 minutes
Date Received	22 May 2013	Date Tested	28 May 2013

**Macro-etch Examination :** The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.**Additional details :** Project Name : MINI SHUTDOWN PTTGC, WPS No.RNC-WPS-023, Filler/Electrode Spec't : ER 347+E347-16

Reported by : 
 (Mr.Keerati Sa-ngoen)
 (Acting) General Manager - Quality Assurance
 Date : 27/5/2013

Approved by : 
 (Mr.Pavaret Preedawiphat)
 Deputy Managing Director
 Date : 28/5/2013

Notes

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LPN PLATE MILL



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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th

13364

Hardness Testing Report

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Page 1 of 1

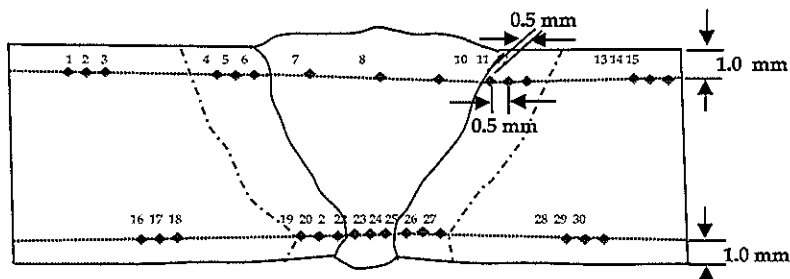
Report No. HV-13-05-042

Revision No. 0

Test Product	PQR No. RNC-PQR-023	Material Characterization	Pipe
Test No.	1474/13 HV	Test Method	ASTM E92-82 (Reapproved 2003) ^{e2}
Project Name	MINI SHUTDOWN PTTGC	Standard of	ASME Section IX : 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	SA312 TP321	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	Ø10" (32 mm Thk.)	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW+SMAW/6G	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	23 May 2013	Cal. Block	No. A60023 No. A62072
Date Tested	27 May 2013	Ambient Temperature	27.9 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)
1	BASE	138.1	16	BASE	153.7	*****		
2	BASE	138.0	17	BASE	154.6			
3	BASE	136.9	18	BASE	154.8			
4	HAZ	139.5	19	HAZ	143.7			
5	HAZ	142.0	20	HAZ	145.7			
6	HAZ	139.0	21	HAZ	152.3			
7	WELD	186.4	22	WELD	194.2			
8	WELD	189.0	23	WELD	203.6			
9	WELD	165.9	24	WELD	210.2			
10	HAZ	144.9	25	HAZ	141.0			
11	HAZ	141.1	26	HAZ	143.4			
12	HAZ	145.9	27	HAZ	142.1			
13	BASE	144.2	28	BASE	153.3			
14	BASE	142.8	29	BASE	153.4			
15	BASE	142.6	30	BASE	154.8			

Additional details : WPS No.RNC-WPS-023, Filler/Electrode Spec't : ER 347+E347-16,

Welder Name/Welder No. : MR.SOMJAI S., MR.KAEN H./W-081,W-079

Reported by :

(Mr.Keerati Sa-ngoan)

(Acting) General Manager - Quality Assurance

Date : 07/06/13

Approved by :

(Mr.Pavaree Preedawiphat)

Deputy Managing Director

Date : 27/07/13

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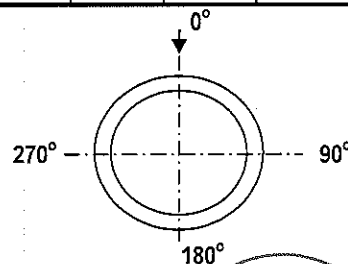
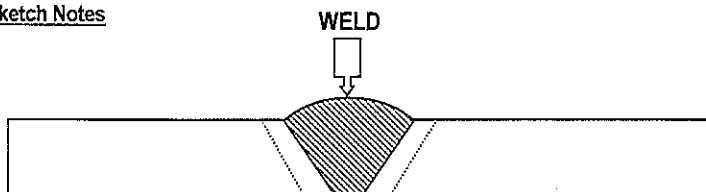
LPN PLATE MILL

FERRITE CONTENT REPORT

Client :	Renew Coils Engineering & Supply Co.,Ltd.	Date :	15-May-13
Project :	Mini Shut Down PTTGC	Report No.:	FR-002
Project No.	PQR	Sheet :	1 OF 1
Acceptance Criteria :-	ASME B 31.3	Analyzer Model :	Feritscope SMP30
WPS No. :	-	Analyzer Serial No. :	021-26950A
Batch / Heat No. :	-	Material Specification / Grade :	A358 TP321

[illegible]

Weld Map/Sketch Notes



	PAE Inspected by	Client By	Approved by
Name	Mr. Phuwadol S.	QA/QC	QA/QC
Signature			
Date	15-May-13	15/05/13	12/05/13

	GLOBAL HEAT TREATMENT [THAILAND] CO., LTD.	
	HEAT TREATMENT REPORT	
Client :	RENEW COILS ENGINEERING & SUPPLY CO., LTD.	Page : 1
Contractor :	GLOBAL HEAT TREATMENT [THAILAND] CO., LTD	Report No. : PW-002
Project :	MINI SHUT DOWN PTTGC 2013	Project No. :
Description :	☒ -PWHT ☐ -Normalize ☐ -Annealing ☐ -Stabilize ☐ -Solution ☐ -Dry-out	
Location :	☒ -Shop ☐ -Field	Recorder No.: EH-05YA054
Method by :	☒ -Electric ☐ -Gas Burner ☐ -Oil burner	Material: SA312 GR. TP321/TP347/SA358TP321
Method Type :	☒ -Local ☐ -Furnace ☐ -Internal Firing	Unit:
Chart Speed :	☒ -25 mm/Hr. ☐ -50 mm/Hr. ☐ -100 mm/Hr.	
Completion Date :	15-05-2013	Procedure No.

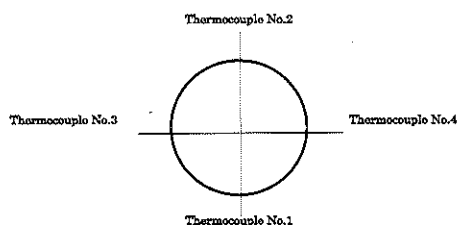
HEAT TREATMENT CONDITION/RESULTS

Description	Loading Temp [1] (°C)	Heating Rate [2] (°C/Hr.)	Holding Temp [3] (Min.-Max.)	Holding Time [4] (Hr.)	Cooling Rate [5] (°C/Hr.)	Unloading Temp [6] (°C)
Request	300	180-200	1080±15	1 HR.	AIR COOLING	300
Actual	300	180	1080	1 HR.	AIR COOLING	300
Request						
Actual						
Request						
Actual						

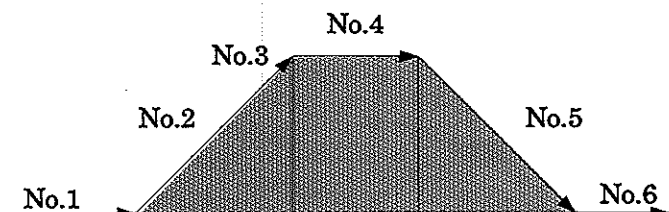
DESCRIPTION

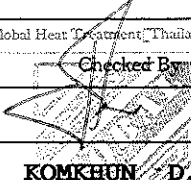
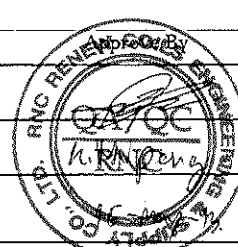
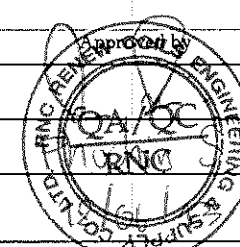
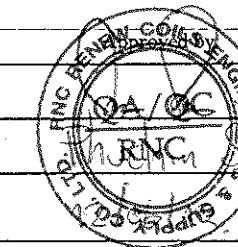
Line No./Dwg.No./Class No./Sht.no./Spool No.	Weld No.	Size(°)	THK.(MM)	Weld Type	T/C No.	Remark
RNC-PQR-023		10"	30	BW	2,3	
Total:		1		Joint.	Total:	10 DB.
Total:				EA.	Total:	Kg.

THERMOCOUPLE LOCATION



HEAT TREATMENT PROFILE



Global Heat Treatment [Thailand] Co., Ltd.				
Signature:				
Name:	KOMKHUN D.			
Date:	15-05-2013			

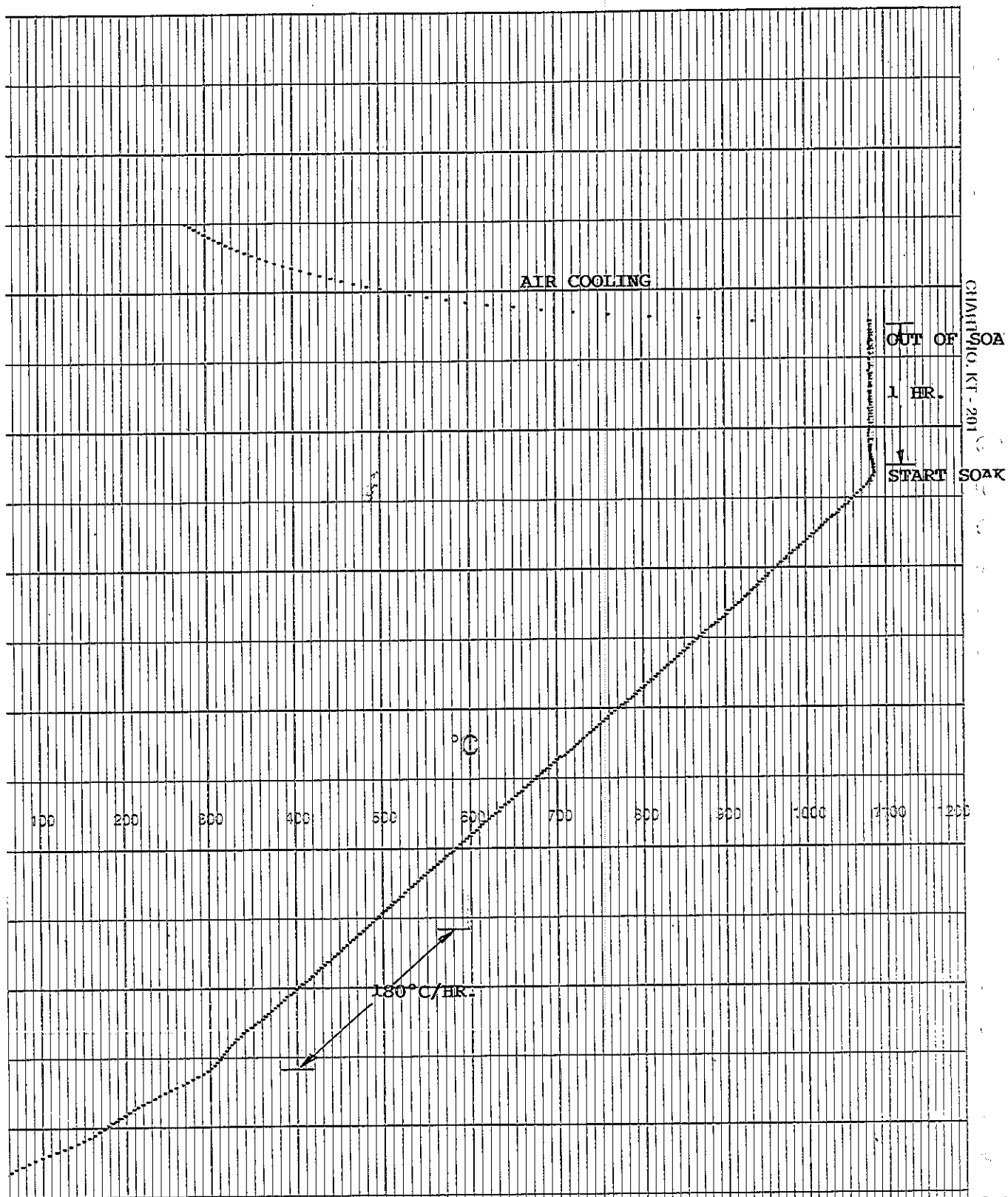


CHART 10, KT - 201

DATE 15-05-2013

CHART NO. PW - 002

RECORDER NO. EH 05YA054

CHART SPEED 25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO., LTD.

PROJECT. MINI SHUT DOWN PITGC 2013

DWG NO. RNC-PQR-023

SIZE 10" THK. 30 mm.

MATERIAL SA312 GR. TP321/TP347/SA358 TP321

START TIME 17.30 PM.

TECHNICIAN MR. CHALERMCHAI NG.

TC

2.

3.

COLOUR

BLACK

SKY BLUE

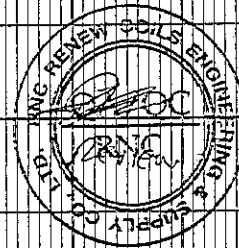
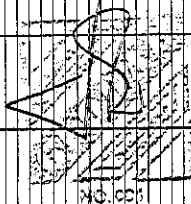


CHART NO. KT - 20

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)682208-9, Fax : 038)682206, SRIRACHA : 038)330220, 330229

Page : 1 Of 1[illegible]

Abnahmeprüfzeugnis 3.1

Inspection certificate 3.1

nach / as per: EN 10204

Nr. / No.: 2-2011-25-1926989

Rev. 0 Seite / page: 1 von/of 1

Bestell-Nr.	Order No.	PO 1030/10.10 - SEA	vom / of	04.10.2010	100049																			
Auftrags-Nr.	Works order	1002107383																						
Lieferschein/Pos./Splitt	Dispatch note/pos./Splitt	2002184537 / 0010 / 900001	vom / of	04.10.2010																				
Prüfgegenstand	Test object	Stabelektrode covered electrode			11958																			
Handelsname	Trade designation	BOEHLER FOX SAS 2-A			Y2053A08																			
Produktkennzeichnung	Marking of product																							
Normbezeichnung	Standard classification	EN 1600: E 19 9 Nb R 32 / AWS A5.4-06: E347-17																						
Abmessung	Dimension	3.20 x 350 mm (.126" x 14")																						
Serie	Lot No.	2123697																						
Liefermenge	Quantity delivered	5006.1 KG (11036 LB)																						
Anforderungen	Requirements																							
Chemische Analyse in % Chemical composition in %								Reines Schweißgut all-weld metal																
Serie Lot No.	C	Si	Mn	P	S	Cr	Mo	Ni	Nb	N2	Cu													
2123697	0.024	0.85	0.71	0.023	0.007	19.69	0.09	9.97	0.35	0.067	0.08													
Zugversuch Tensile Test								Probenvorbereitung Specimen preparation																
	nach according to	ASTM E8						AWS B4.0																
	Prüftemp. T Test temp. °C	Streckgrenze Reh Yield point MPa	Dehngrenze Rp Yield strength 0.2% MPa 1.0%					Zugfestigkeit Rm Tensile strength MPa	Dehnung A Elongation % Lo=4d	Einschr. Z Reduction %	Bemerkung Remarks													
Minimum	20							520	30															
Maximum																								
Kerbschlagv. Impact Test								Probenvorbereitung Specimen preparation																
	nach according to	ASTM E23						AWS B4.0																
	Prüftemp. T Test temp. °C	Kerbschlagarbeit Mindest. KV Absorbed energy minimum values				Mittelwert Average J		Lat. Breitg. Lateral expansion mm	Krist. Bruchanteil Shear fracture %		Bemerkung Remarks													
	20°C (68°F)	55																						

Bemerkung
Remarks

The product BOEHLER FOX SAS 2-A meets the requirements of the filler metal specification ASME, sec.II, part C, AWS A5.4-06, classification E347-17 when tested in accordance with that specification. Produced according to AWS A5.01, class C3

Ort / Town
Kapfenberg

Datum / Date
29.10.2010

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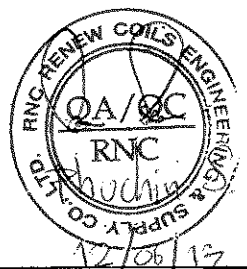
Abnahmebeauftragter
Authorized representative
Winkler

Böhler Schweistechnik Austria GmbH
Böhler-Welding-Str. 1
A-8605 Kapfenberg
Internet: www.boehler-welding.at

Tel: +43 (0) 3862-301-0
Fax: +43 (0) 3862-301-45-193
E-Mail: postmaster.bega@bega.at

Registriergericht: Leoben, FN 77052 M
ATU 27116607
DVR 0862348

Deutsche Bank AG (BLZ 19100); Nr. 31486000
Swiftcode: DEUTAT33
IBAN: AT37 1910 0000 3148 6000



WERKSZEUGNIS 3.1

INSPECTION CERTIFICATE

nach / as per: EN 10204

Nr. / No.: 2-2011-03-1993640

Rev. 0 Seite / page: 1 von/ of 1

Bestell-Nr.	Order No.	PO 1056/10.11-airfreight	vom / of	14.02.2011	100048											
Auftrags-Nr.	Works order	1002110557														
Lieferschein/Pos./Spfitt	Dispatch note/pos./Spfitt	2002188963 / 0010 / 900001	vom / of	14.02.2011												
Prüfgegenstand	Test object	Schweißstab welding rod			71409											
Handelsname	Trade designation	BOEHLER SAS 2-IG			2804ENK2											
Produktkennzeichnung	Marking of product															
Normbezeichnung	Standard classification	EN ISO 14343-A - W 19 9 Nb / AWS A5.9-06: ER347														
Abmessung	Dimension	2.40 x 1000 mm (.094" x 39")														
Charge	Heat No.	97694														
Liefermenge	Quantity delivered	300 KG														
Anforderungen	Requirements															
Chemische Analyse in % Chemical composition in %		Stab: rod														
Charge Heat No.	C	Si	Mn	P	S	Cr	Mo	Ni	Nb	Cu						
97694	0.044	0.47	1.80	0.023	0.007	19.63	0.21	9.10	0.61	0.13						
Zugversuch Tensile Test	nach according to	EN 10002-1		Probenvorbereitung Specimen preparation		EN 876										
	Prüftemp. T Test temp. °C	Streckgrenze Reh Yield point MPa	Dehngrenze Rp Yield strength 0.2% MPa 1.0%	Zugfestigkeit Rm Tensile strength MPa	Dehnung A Elongation % Lo=5d	Einschr. Z Reduction %	Bemerkung Remarks									
Minimum	20		450	600	30											
Maximum																
Kerbschlagv. Impact Test	nach according to	EN 10045-1		Probenvorbereitung Specimen preparation		EN 875 VWT 0/b										
	Prüftemp. T Test temp. °C	Kerbschlagarbeit Mindest. KV Absorbed energy minimum values J		Mittelwert Average J	Lat. Breitg. Lateral expansion mm	Krist. Bruchanteil Shear fracture %	Bemerkung Remarks									
	20	100														
	-196	32														

Bemerkung
Remarks

Ort / Town
Kapfenberg

Datum / Date
15.02.2011

Dieses Zeugnis wurde maschinell erstellt und gilt ohne Unterschrift.
This certificate was issued by DP-equipment and does not require signature.

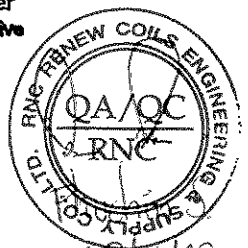
Abnahmebeauftragter
Authorized representative
Winkler

Böhler Schweißtechnik Austria GmbH
Böhler-Welding-Stk. 1
A-8805 Kapfenberg
Internet: www.boehler-welding.at

Tel: +43 (0) 3862-301-0
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